Anvitha Reddy

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PROFILE SUMMARY

AI Engineer building production-grade Generative AI systems at Advanced Micro Devices — specializing in RAG pipelines, Text-to-SQL, and LLM orchestration at enterprise scale. Previously engineered multilingual NLP systems at CornerstoneX spanning 50+ languages with CUDA-optimized inference. Core stack: Python, LangChain, OpenAI API, Anthropic Claude API, Pinecone, and MCP (Model Context Protocol).

WORK EXPERIENCE

AI Engineer — Advanced Micro Devices (AMD) | ZenBot PM Digital TwinAug 2025 – Present

- Delivered enterprise-scale Text-to-SQL access across 4 data sources (JIRA, JAMA, ATM, Test Reports) by architecting a two-model LLM cascade (OpenAI + Anthropic Claude) for intent classification and SQL generation, enabling non-technical stakeholders to self-serve complex data queries with zero SQL knowledge.
- Reduced LLM hallucination rate by implementing AST-level SQL validation (sqlglot) with an auto-repair loop using live INFORMATION_SCHEMA context; optimized API cost and latency by routing requests through a lightweight intent classifier before the generation model, maintaining production-quality output while lowering inference spend.
- Built a production RAG pipeline over an enterprise corpus (WHQL, IV, IQE, reliability and declaration reports) using Pinecone, Weaviate, and FAISS for vector storage with semantic retrieval and disambiguation blocks, eliminating document-type conflation across 5+ report categories.
- Engineered high system uptime on on-premises infrastructure by implementing circuit breakers, TTL-based response caching, and graceful degradation fallbacks at every layer, achieving zero cascading failures during peak query load.
- Drove 70%+ test coverage by establishing a CI/CD pipeline with ruff lint, mypy type checking, and unit/integration test separation enforced on every PR; achieved zero-code domain onboarding via a config-driven JSON architecture.

Machine Learning Engineer — CornerstoneX **Nov 2024 – Aug 2025**

- Spearheaded a multilingual conversational AI platform supporting 50+ languages by architecting end-to-end NLP pipelines on Whisper, Hugging Face Helsinki/M2M100, and ElevenLabs with CUDA-optimized inference, boosting global customer engagement by 300% and reducing customer support tickets by 40%.
- Delivered over 90% translation accuracy and sub-2-second latency at scale; integrated Metahuman avatars to enhance AI-human interaction, driving a 40% surge in enterprise demo requests.
- Engineered a CNN-based ASL Gesture Recognition System achieving 95%+ accuracy across 1,000+ gestures at 15 FPS, earning pilot approval from the U.S. Department of Homeland Security.
- Built a voice-controlled navigation assistant with 85%+ precision on multi-intent commands, improving task efficiency by 40% for users with mobility challenges; contributed to the AI project that earned CornerstoneX selection for the NVIDIA Inception Program and was presented at NVIDIA GTC 2025.

EDUCATION

Christ University **Aug 2021 – May 2025**
B.Tech in Computer Science (Data Science) | Bengaluru, KA GPA: 3.5/4.0

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java

AI/ML & GenAI: Large Language Models (LLM), Retrieval-Augmented Generation (RAG), Text-to-SQL,

Natural Language Processing (NLP), Prompt Engineering, LLM Orchestration, Agentic AI / Multi-Agent Systems, Intent Classification, Semantic Search, Embedding Models, Deep Learning, Computer Vision, Speech Recognition, Text-to-Speech, Machine Translation

LLM APIs: OpenAI API, Anthropic Claude API

Vector Databases: Pinecone, Weaviate, FAISS

Frameworks & Libraries: LangChain, LlamaIndex, Hugging Face, Whisper, M2M100, Helsinki-NLP, ElevenLabs, Ollama, sqlglot, FastAPI

Tools & Platforms: n8n, CUDA, Snowflake, MCP (Model Context Protocol), Git, Linux, Docker, On-premises GPU infrastructure

Engineering Practices: CI/CD, ruff, mypy, Unit Testing, Integration Testing, Circuit Breakers, Config-Driven Architecture, TTL Caching, Graceful Degradation

CERTIFICATIONS AND TRAINING

Tata Group — Data Visualization (Forage Virtual Experience) **2023**
Completed a job simulation focused on data storytelling and executive-level dashboard design for business stakeholders.